Chapter 8: Price Level Determination under Sustainable Fiscal Policy

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1 Overview

We introduce money in the most simple way to determine inflation. Even if the model is limited, it produces insights that also apply in more general settings.

When the central bank decides that it wants to set, for example, 2% inflation, a fundamental question arises: how does it want to do this? Milton Friedman suggested that the best way to guard price stability was to have the money supply increase by a fixed amount every year. A much more common approach by central banks is to set the nominal interest rate in a way that promotes price stability. The question we ask here is: if the central bank selects a particular path for its policy instruments – the money supply or the nominal interest rate – what happens to inflation? Key conclusions are:

i. Inappropriate monetary policy rules can create "indeterminacy," meaning that the movements in inflation could depend on totally random things, such as the color of the tie of the Chairman of the Federal Reserve.

ii. We will show a simple result about the appropriate "intermediate target" of the central bank – i.e. whether it should set the interest rate or money supply depends upon the relative sizes of fundamental shocks in the economy. The result helps when thinking about why most central banks today use the interest rate as the intermediate target when setting monetary policy.

2 Monetary and Fiscal Policy: Defining an Equilibrium

A representative household maximizes:

$$\max_{C_t, M_t, B_t} E_0 \sum_{t=0}^{\infty} \beta^t \left[u(C_t) + \chi \left(\frac{M_t}{P_t} \right) \right] \xi_t$$

s.t.

$$M_t + B_t = (1 + i_{t-1}) B_{t-1} + M_{t-1} + P_t Y_t - P_t T_t - P_t C_t \tag{1}$$

where B_t- is one-period nominal risk-free bond, Y_t- is income, C_t- is consumption, P_t- is the price level. M_t- is money, T_t- is taxes, ξ_t- is a preference shock.

We assume that there is satiation of the real monetary balance at $\bar{m}$, so that $\chi_m(\frac{M_t}{P_t}) = 0$ for $m_t \geq \bar{m}$, where χ_m is the partial derivative of the function $\chi(\frac{M}{P})$ with respect to $\frac{M}{P}$. For now, however, we will suppose that $m < \bar{m}$, ignoring this possibility for the time being, but we will get back to it later as it will be important when we consider the zero lower bound.

We can derive the first-order conditions by solving the Lagrangian, assuming $m < \bar{m}$:

$$\mathcal{L}_0 = E_0 \sum_{t=0}^{\infty} \beta^t \Bigg\{ \left[u(C_t) + \chi(\frac{M_t}{P_t}) \right] \xi_t + \phi_t [M_t + B_t - (1 + i_{t-1}) B_{t-1} - M_{t-1} - P_t Y_t + P_t T_t + P_t C_t] \Bigg\}$$

The first-order conditions (FOC) of this Lagrangian are:

$$\begin{aligned}
\frac{\partial \mathcal{L}_0}{\partial C_t} &= \beta^t u_c(C_t) \xi_t + \beta^t \phi_t P_t = 0 \\
\frac{\partial \mathcal{L}_0}{\partial B_t} &= \beta^t \phi_t - \beta^{t+1} (1 + i_t) E_t \phi_{t+1} = 0 \\
\frac{\partial \mathcal{L}_0}{\partial M_t} &= \beta^t \chi_m(\frac{M_t}{P_t}) \frac{1}{P_t} \xi_t + \beta^t \phi_t - \beta^{t+1} E_t \phi_{t+1} = 0
\end{aligned}$$

where $u_c(C)$ denotes the partial derivative of the function $u(.)$ with respect to C.

We can use the first FOC to substitute out for ϕ_t in the second equation, obtaining the ***Consumption Euler Equation***:

$$\frac{1}{1 + i_t} = \beta E_t \frac{u_c(C_{t+1})}{u_c(C_t)} \frac{\xi_{t+1}}{\xi_t} \frac{P_t}{P_{t+1}} \tag{2}$$

We can similarly substitute out for ϕ_t in the last FOC and use (2) to yield the ***Money Demand Equation***:

$$\frac{\chi_m(\frac{M_t}{P_t})}{u_c(C_t)} = \frac{i_t}{1+i_t} \tag{3}$$

Defining $V_t = M_{t-1} + (1+i_{t-1})B_{t-1}$, then the transversality condition can be written as:

$$\lim_{T\to\infty} E_t Q_{t,T} V_T = 0 \tag{4}$$

where $Q_{t,T}$ is the stochastic discount factor and $Q_{t,T} \equiv \beta^{T-t}\frac{u_c(C_T)\xi_t P_T}{u_c(C_t)\xi_t P_t}$. Both the transverality condition and the stochastic discount factor can be derived in the same way as we did in last chapter. Note that the consumption Euler Equation now refers to the pricing of a one-period risk-free *nominal* bond, i.e. a one-period bond that pays $1+i_t$ units of *dollars* with certainty in period $t+1$.

The resource constraint is:

$$Y_t = C_t + G_t \tag{5}$$

Use this to rewrite the budget constraint of the household in terms of the government budget constraint:

$$M_t + B_t = (1+i_{t-1})B_{t-1} + M_{t-1} + P_t G_t - P_t T_t$$

or:

$$V_{t+1} = (1+i_t)\left(V_t - \frac{i_t}{1+i_t}M_t + P_t G_t - P_t T_t\right) \tag{6}$$

Fiscal policy is now defined as choice of debt and government spending:

$$T_t = T(\Pi_t, Y_t, \xi_t,) \tag{7}$$

$$G_t = G(\Pi_t, Y_t, \xi_t,) \tag{8}$$

defining the sequences $\{T_t\}$ and $\{G_t\}$. For a given monetary policy, the choice of $T(.)$ and $G(.)$ defines the value of V_{t+1}. While fiscal policy specifies spending and overall debt (V_{t+1}) for a given monetary policy, monetary policy specifies how V_{t+1} is split between money and bonds via open market operations, given fiscal policy, i.e.:

$$M_t = M(\Pi_t, Y_t, \xi_t) \tag{9}$$

We can alternatively specify monetary policy by an interest rate reaction function:

$$i_t = i(\Pi_t, Y_t, \xi_t), \tag{10}$$

where the interpretation is that the central bank does open market operations so as to target the interest rate via the money demand equation. We have not here directly written out the budget constraint of the Central Bank vs the Treasury, but this is a straightforward extension.

Definition 1. *An equilibrium consistent with* risk-free nominal government debt *is a collection of stochastic processes* $\{V_{t+1}, T_t, G_t, M_t, i_t, C_t, P_t\}$ *for* $t \geq 0$ *that satisfy (2), (3), (4), (5) and (6) given a process for* $\{Y_t, \xi_t\}$ *and a policy specification given by (7) and (8) for fiscal policy and (9) OR (10) for monetary policy, and an initial condition* V_0.

Let us define a sustainable fiscal policy as:

Definition 2. *A sustainable fiscal policy is a specification for* $T(.)$ *and* $G(.)$*, with monetary policy given by (9) (or (10)), given* V_0*, so that (4) and (6) are satisfied for any path of* $\{i_t, P_t, M_t, \xi_t, Y_t\}$.

The following proposition is very useful and is analogous to that of the Ricardian Equivalence we have already proved. It says that we can analyze the determination of prices without worrying about fiscal policy, if fiscal policy is sustainable.

Proposition 1. *Assume* $T(.)$ *and* $G(.)$ *such that fiscal policy is sustainable. Then the equilibrium determination of* $\{M_t, i_t, C_t, P_t, G_t\}$ *is independent of the specification of the function* $T_t = T(\Pi_t, Y_t, \xi_t)$ *and public debt* V_{t+1}.

Proof. Equilibrium conditions for $\{i_t, P_t, C_t, M_t, G_t\}$ given $\{Y_t, \xi_t\}$ are given by:

$$\frac{1}{(1+i_t)} = \beta E_t \frac{u_c(Y_{t+1} - G_{t+1}(.))}{u_c(Y_t - G_t(.))} \frac{\xi_{t+1}}{\xi_t} \frac{P_t}{P_{t+1}} \tag{11}$$

$$\frac{\chi_m\left(\frac{M_t}{P_t}\right)}{u_c(Y_t - G_t(.))} = \frac{i_t}{1+i_t} \tag{12}$$

$$M_t = M(\Pi_t, \xi_t) \text{ or } i_t = i(\Pi_t, Y_t, \xi_t)$$

Observe that none of these equations depend on the function $T(\Pi_t, Y_t, \xi_t)$ or public debt V_{t+1}. A candidate equilibrium $\{i_t, P_t, C_t, M_t, G_t\}$ given $\{Y_t, \xi_t\}$ is thus valid as long as the choice of $T(.)$ satisfies:

$$V_{t+1} = (1+i_t)\left(V_t - \frac{i_t}{1+i_t} M_t + P_t G_t - P_t T_t\right)$$

$$\lim_{T\to\infty} \beta^T Q_{t,T} V_T = 0$$

which it does by the definition of sustainable policy. □

The assumption of sustainable policy puts strong restrictions on taxes. The point is that there are several types of policies $T(.)$ that satisfy the two equations above, implying various different evolutions of debt over time, leading to the validity of the celebrated conclusion that the exact timing of taxes is irrelevant (Ricardian Equivalence). More importantly, we can determine the price level independently of tax policy and the level of public debt. The point is not that taxes are always irrelevant for prices. Indeed, we will analyze cases when taxes and public debt may play an important role. Before we do that, however, it is of interest to contemplate the more limited question where fiscal policy plays no role, which provides several important insights.

3 Price Level Determination with Sustainable Fiscal Policy

In this section, we assume that fiscal policy is sustainable. Consider the following example of a sustainable fiscal policy.

Fiscal policy: $\{V_t\}$ and $\{G_t\}$ are bounded exogenous sequences. Taxes adjust endogenously so as to ensure that the government budget constraint is satisfied. In equilibrium, then, taxes are set to yield:

$$T_t = \frac{V_t}{P_t} - \frac{1}{1+i_t}\frac{V_{t+1}}{P_t} - \frac{i_t}{1+i_t}\frac{M_t}{P_t} + G_t$$

It is easy to show that this is an example of sustainable fiscal policy.

It follows from Proposition 1 that the price level does not depend on taxes or public debt. This is what allows us to abstract from fiscal policy when we talk about how prices are determined. For simplicity, we assume that $G_t = 0$, as nothing depends upon this assumption. It is useful to redefine the equilibrium in terms of the following variables. We define:

$$\mu_t \equiv \frac{M_t}{M_{t-1}}$$
$$m_t \equiv \frac{M_t}{P_t}$$
$$\Pi_t \equiv \frac{P_t}{P_{t-1}}$$

It follows from these definitions that:

$$m_t = m_{t-1}\mu_t\Pi_t^{-1}$$

Monetary policy: Defined by one of the two functions:

$$\mu_t = \mu\left(Y_t, \Pi_t, \xi_t\right) \tag{13}$$

or:

$$i_t = i\left(Y_t, \Pi_t, \xi_t\right) \tag{14}$$

Assuming $m_t < \bar{m}$, we can then summarize the model with the following system of equations:

$$\frac{1}{1+i_t} = \beta E_t \frac{u_c(Y_{t+1})}{u_c(Y_t)} \frac{\xi_{t+1}}{\xi_t} \Pi_{t+1}^{-1} \tag{15}$$

$$\frac{\chi_m(m_t)}{u_c(Y_t)} = \frac{i_t}{1+i_t} \tag{16}$$

$$m_t = m_{t-1}\mu_t \Pi_t^{-1} \tag{17}$$

together with (13) or (14). The government budget constraint will simply back out the T_t needed to implement a given equilibrium.

Does a given money supply rule or interest rate rule give a unique solution? We will proceed with the following common strategy to characterize dynamics in these notes, 1) find a steady state (constant solution), 2) linearize around it via first-order Taylor expansion, 3) solve the set of linearized equations.

We start with the steady state. Consider a steady state in where all variables are constant, i.e. $\Pi_t = \bar{\Pi}$, $m_t = \bar{m}$, $i = \bar{\imath}$ and $\mu_t = \bar{\mu} > 0$. Using equation (17), we get:

$$\bar{\mu} = \bar{\Pi}$$

i.e. prices grow at the same rate as money. This is a basic quantity theory of money: inflation increases proportionally with the growth rate of money supply. Some evidence is shown in Robert Lucas's Nobel lecture (Figure 1).

Using instead the consumption Euler Equation (15) and setting variables constant, we get:

$$1 + \bar{\imath} = \beta^{-1}\bar{\Pi}$$

which means:

$$1 + \bar{\imath} = \beta^{-1}\bar{\Pi} = \beta^{-1}\bar{\mu},$$

implying the simple empirical relationship shown in Figure 2, taken from Schmitt-Grohe and Uribe (2016).

Let assume the following functional forms: $u(C) = \frac{C^{1-\sigma^{-1}}}{1-\sigma^{-1}}$ and $\chi(\frac{M}{P}) = \frac{(\frac{M}{P})^{1-\nu}}{1-\nu}$. Then, from money demand equation (16), we derive:

$$\bar{m} = \left(\frac{\bar{\mu}\beta^{-1}}{\bar{\mu}\beta^{-1} - 1}\right)^{\frac{1}{\nu}} \bar{Y}^{\frac{\sigma}{\nu}},$$

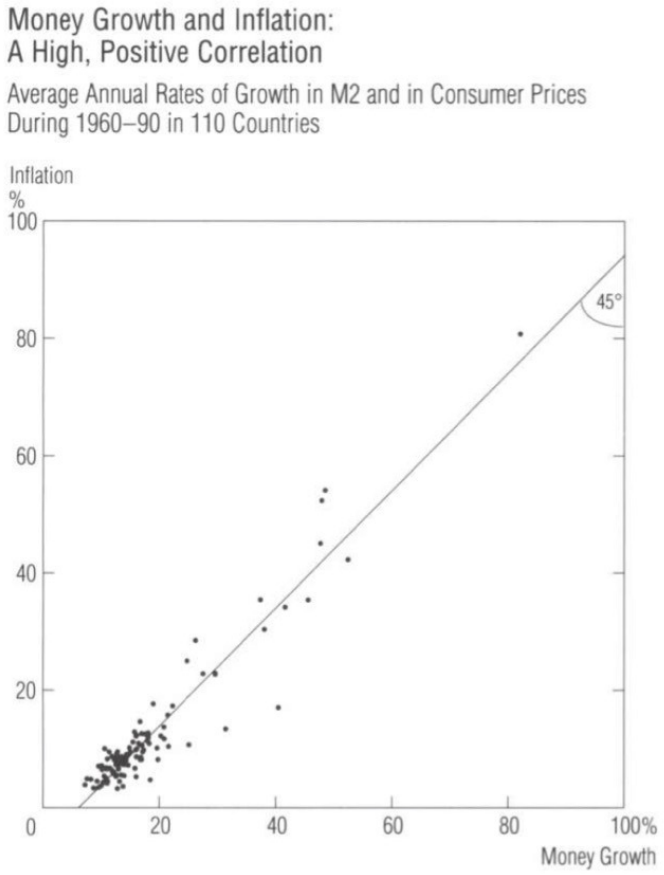


Figure 1: Growth Rate of Money and Inflation

suggesting that $\bar{m}$ is generally decreasing in $\bar{\mu}$: a higher nominal interest rate reduces real money balance (we do have evidence of this from high inflation episodes).

Define the variables by deviation from the steady state: $\hat{Y}_t \equiv \frac{Y_t - \bar{Y}}{\bar{Y}}$, $\hat{C}_t \equiv \frac{C_t - \bar{C}}{\bar{C}}$, $\hat{m}_t \equiv \frac{m_t - \bar{m}}{\bar{m}}$, $\hat{\imath}_t \equiv \frac{i_t - \bar{i}}{1+\bar{i}}$ [1], $\pi_t \equiv \Pi_t - 1$ (so $\bar{\Pi} = 1$). Linearizing (15), assuming for simplicity that $G_t = 0$ and the functional forms above, yields:

$$\hat{\imath}_t = \hat{r}_t + E_t \pi_{t+1} \tag{18}$$

where $\hat{r}_t \equiv \sigma^{-1} E_t(\hat{Y}_{t+1} - \hat{Y}_t) - E_t \hat{\xi}_{t+1} + \hat{\xi}_t$. Written in this form, this equation is often referred to as the Fisher equation. It can be shown that $\hat{r}_t$ also corresponds

[1] This comes from a Taylor expansion.

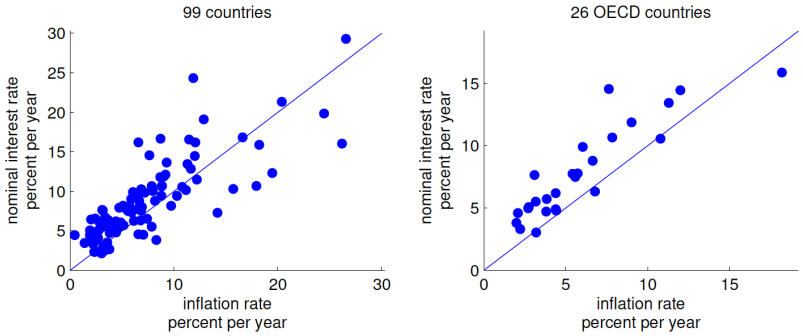


Figure 2: Inflation and Interest Rates

to the price of a one-period risk-free real bond (we saw this in the lecture on fiscal policy). Meanwhile, linearizing the money demand (16) yields:

$$\hat{m}_t = \eta_y \hat{Y}_t - \eta_i \hat{\imath}_t + \epsilon_t^{md} \tag{19}$$

where $\eta_y \equiv \nu^{-1}\sigma^{-1}$ and $\eta_i \equiv \nu^{-1}\frac{\beta}{1-\beta}$ (since $1 + \bar{\imath} = \beta^{-1}\bar{\Pi}$ and $\bar{\Pi} = 1$). Linearizing the identity (20) yields:

$$\hat{m}_t = \hat{m}_{t-1} + \hat{\mu}_t - \pi_t, \tag{20}$$

while the policy rule is:

$$\hat{\mu}_t = \mu_\pi \pi_t + \varepsilon_t^\mu \quad \text{or} \quad \hat{\imath}_t = \phi_\pi \pi_t + \varepsilon_t^i \tag{21}$$

where ε_t^μ and ε_t^i are exogenous sequences.

We can now define an approximate equilibrium:

Definition 3. *An approximate equilibrium under sustainable fiscal policy is a collection of stochastic processes* $\{i_t, \pi_t, \hat{m}_t, \hat{\mu}_t\}$ *for* $t \geq 0$ *that satisfies (18), (19), (20) and (21), given an exogenous process for* $\{\hat{r}_t, \hat{Y}_t\}$.

3.1 The Great Inflation and the Problem of Self-Fulfilling Expectations

3.1.1 Sargent and Wallace's interest rate indeterminacy result

We are now in a position to address the question posed at the beginning. What kind of strategy would work to implement an inflation target of, say, 2 percent? Consider first a strategy focused on interest rate setting. One natural guess, based upon our expression for the steady-state nominal interest rate in (3) is to simply set $it = \bar{\imath}$. Since this corresponds to the steady state, the hope would then be that the economy would ultimately settle on the inflation rate consistent with this steady state. The first result we show here, based on Sargent and Wallace (1975), shows that this strategy would lead to indeterminacy, i.e. are arbitrary fluctuations in the price level are possible.

If monetary policy is formulated via an interest rate rule, then we do not need to keep track of the money supply. To see this, note that equation (18) and the interest rate rule in (21) are two equations in $\hat{\imath}_t$ and π_t, so that, given $\hat{\imath}_t$ and π_t (jointly determined by (18) and the second equation in (21)), then equations (19) and (20) only matter in order to pin down $\{\hat{m}_t, \hat{\mu}_t\}$. This leads us to the Sargent-Wallace result:

Proposition 2 (Sargent and Wallaces' Interest Rate Indeterminacy Result)**.** *If monetary policy is set by an exogenous sequence $\{i_t\}$, the price level is indeterminate.*

This result is simply proved by an example that depends on an arbitrary "sunspot variable," which satisfies the equations of the model. Suppose:

$$\hat{\imath}_t = \varepsilon_t^i \tag{22}$$

where $\{\epsilon_t^i\}$ is some arbitrary exogenous sequence. One possible solution is then:

$$\pi_t = \hat{\imath}_{t-1} - \hat{r}_{t-1} + \omega_t \tag{23}$$

where ω_t can be any random sunspot variable such that $E_t\omega_{t+1} = 0$. To verify that this is indeed a solution, just iterate equation (23) one period forward so that: $\pi_{t+1} = \hat{\imath}_t - \hat{r}_t + \omega_{t+1}$. Now take expectation on both sides: $E_t\pi_{t+1} = \hat{\imath}_t - \hat{r}_t + E_t\omega_{t+1} = \hat{\imath}_t - \hat{r}_t$. This shows that the proposed equilibrium (23) satisfies both (18) and (21) and thus is a valid equilibrium (recall that these two equations are sufficient to pin down equilibria). Hence, according to Sargent and Wallace's argument, interest rate targeting is not a good idea since an interest rate target doesn't rule out self-fulfilling expectations. One theory suggests that the great inflation in the 1970s occurred due to self-fulfilling expectations.

The Sargent and Wallace result is incredibly useful to organize thinking about monetary policy. To me, the most important takeaway is this general observation: suppose that you just pin down a path for the Federal Funds Rate (for example, via so-called "forward guidance"). The Sargent and Wallace result teaches us that, in general, there is in fact an infinite number of possible equilibria that are consistent with this policy specification. This is an observation that not only holds in the simple example above, but also much more generally in rational expectation models.

3.1.2 Sargent and Wallace money supply policy regime

While setting the interest rate equal to its steady state value was one natural guess for policy to implement a given inflation target, another natural alternative is to do the same with the money supply, i.e. using equation (3) set $\mu_t = \Pi^*$. Based on the previous result, one might think that a similar indeterminacy result would also apply in this case – after all, either the money supply or interest rate as the "policy instrument" have often been suggested as two alternative choices for monetary policy, of "intermediate targets." Sargent and Wallace, however, showed – as we confirm below – that an exogenous money supply rule does instead lead to determinacy and thus is the optimal policy instrument.

Proposition 3 (Sargent and Wallace's Money Determinacy Result)**.** *Suppose monetary policy is set by an exogenous sequence for the money supply* $\{\hat{\mu}_t\}$*. Then, there is a unique bounded rational expectation equilibrium for the price level.*

To prove this, consider the following process for the money supply:

$$\hat{\mu}_t = \varepsilon_t^{\mu} \tag{24}$$

and suppose that, for simplicity, $E_t \varepsilon_{t+j}^{\mu} = 0$ for $j > 0$ (this is not strictly needed, but the generalization does not yield much extra insight). We can solve the model by plugging the Fisher equation (18) in the money demand equation (19) to get:

$$\hat{m}_t = \eta_y \hat{Y}_t - \eta_i \hat{r}_t - \eta_i E_t \pi_{t+1} + \epsilon_t^{md}$$

Now compute inflation from (20), iterate forward and apply expectation to both sides:

$$\begin{aligned} \pi_t &= \hat{m}_{t-1} + \hat{\mu}_t - \hat{m}_t \\ E_t \pi_{t+1} &= \hat{m}_t + E_t \hat{\mu}_{t+1} - E_t \hat{m}_{t+1} = \hat{m}_t - E_t \hat{m}_{t+1} \end{aligned}$$

Replace expected inflation in the equation derived just above:

$$\begin{aligned}\hat{m}_t &= \eta_y \hat{Y}_t - \eta_i \hat{r}_t - \eta_i E_t \pi_{t+1} + \epsilon_t^{md} \\ &= \eta_y \hat{Y}_t - \eta_i \hat{r}_t - \eta_i \left(\hat{m}_t - E_t \hat{m}_{t+1}\right) + \epsilon_t^{md}\end{aligned}$$

Rearrange:

$$\hat{m}_t = \left(\frac{\eta_i}{1+\eta_i}\right) E_t \hat{m}_{t+1} + \left(\frac{1}{1+\eta_i}\right) \left(\eta_y \hat{Y}_t - \eta_i \hat{r}_t + \epsilon_t^{md}\right)$$

Iterating forward, we can solve the equation and get:

$$\hat{m}_t = \left(\frac{\eta_i}{1+\eta_i}\right)^T E_t \hat{m}_{t+T} + \frac{1}{1+\eta_i} E_t \sum_{j=0}^{T-1} \left(\frac{\eta_i}{1+\eta_i}\right)^j \left(\eta_y \hat{Y}_{t+j} - \eta_i \hat{r}_{t+j} + \epsilon_{t+j}^{md}\right),$$

which yields a unique bounded solution if $\frac{\eta_i}{1+\eta_i} < 1$ (note that this is always the case since $\eta_i = \nu^{-1} \frac{\beta}{1-\beta} > 0$):

$$\hat{m}_t = \frac{1}{1+\eta_i} E_t \sum_{j=0}^{\infty} \left(\frac{\eta_i}{1+\eta_i}\right)^j \left(\eta_y \hat{Y}_{t+j} - \eta_i \hat{r}_{t+j} + \epsilon_{t+j}^{md}\right) \tag{25}$$

This is Sargent and Wallace's determinacy result. A policy with an exogenous sequence for the money supply $\{\hat{\mu}_t\}$ yields a unique bounded solution, in sharp contrast to the indeterminacy of the interest rate rule. They argued that this means that the money supply, not the interest rate, is the optimal policy instrument. This result, among others (especially arguments by Milton Friedman), led the Federal Reserve to target the money supply in the 1980s, under Volcker, in order to bring down inflation.

3.1.3 Taylor rules

The general consensus seems to be that the Fed's experience with targeting money supply in the early 1980s was not a success. The key reason was that it resulted in large fluctuations in the interest rate that were not justified by variability in inflation. This ultimately led the Fed to abandon money targeting and today the Federal Funds Rate remains the main measure of the Fed's policy stance. But did Sargent and Wallace's result not show that this leads to instability? While the key insight has stood the test of time and remains one of the most fundamental in monetary economics, subsequent research qualified their conclusion, at least as it is applied to monetary policy.

McCallum (1982) [1] came up with an important extension to the Sargent-Wallace result. He pointed out that their analysis assumed an interest rate rule that specifies an *exogenous path* for the short-term nominal interest rate. In practice, however, this is not how monetary policy is done when interest rates are the policy instrument. Instead, interest rates respond to the endogenous variables of the model. In particular, they respond to the level of inflation as in equation (21), which has been known as the Taylor rule. Consider the interest rate reaction function in (21). We now have the following result:

Proposition 4 (Taylor Principle)**.** *Suppose that monetary policy is set so that* $i_t = \phi_\pi \pi_t + \epsilon_t$*. Then, there is a unique bounded rational expectation equilibrium for the price level if* $|\phi_\pi| > 1$*.*

To prove this proposition, substitute (21) into (18) to obtain: $\hat{r}_t + E_t\pi_{t+1} = \phi_\pi \pi_t + \varepsilon_t^i$. Solving for π_t yields:

$$\begin{aligned} \pi_t &= \frac{1}{\phi_\pi} E_t \pi_{t+1} + \frac{1}{\phi_\pi}(\hat{r}_t - \varepsilon_t^i) \\ &= \left(\frac{1}{\phi_\pi}\right)^T E_t \pi_{t+T} + \sum_{j=0}^{T-1} \left(\frac{1}{\phi_\pi}\right)^{j+1} E_t\left(\hat{r}_{t+j} - \varepsilon_{t+j}^i\right) \end{aligned}$$

which yields a unique bounded solution:

$$\hat{\pi}_t = \sum_{j=0}^{\infty} \left(\frac{1}{\phi_\pi}\right)^{j+1} E_t\left(\hat{r}_{t+j} - \varepsilon_{t+j}^i\right) \tag{26}$$

if $|\phi_\pi| > 1$. That $|\phi_\pi| > 1$ is often referred to as the **Taylor principle**, after the economist John Taylor.

One theory of the great inflation of the 1970s (from Clarida, Galí, and Gertler (1999) – see Figure 3 below, where $\beta = \phi_\pi$) is that $|\phi_\pi| < 1$ pre-Volcker. Hence, what was needed to bring down inflation and achieve stability was not really a monetary target, but instead that the interest rate be raised more than one-to-one with respect to inflation. This, however, is not the only way in which interest rate targeting can be conducted (i.e. reacting to changes in inflation), which brings us to – of all places – Sweden.

3.1.4 Wicksellian price level target

"So long as prices remain unaltered the central bank's rate of interest is to remain unaltered. If prices rise, the rate of interest is to be raised, and if prices fall, the rate of interest is to be lowered" – Knut Wicksell, 1898.

An alternative way of controlling the price level, suggested by the Swedish economist Wicksell more than a hundred years ago, is to increase the interest

	π^*	β	γ	ρ	p
Pre-Volcker	4.24	0.83	0.27	0.68	0.834
	(1.09)	(0.07)	(0.08)	(0.05)	
Volcker-Greenspan	3.58	2.15	0.93	0.79	0.316
	(0.50)	(0.40)	(0.42)	(0.04)	

Standard errors are reported in parentheses. The set of instruments includes four lags of inflation: output gap, the federal funds rate, the short-long spread, and commodity price inflation.

Figure 3: The Taylor Principle

rate when the price level is above a target level, which we call P_t^*. A formalization of this idea is to specify the policy rule as:

$$\hat{\imath}_t = \phi_P \hat{P}_t + \varepsilon_t^i \tag{27}$$

where $\hat{P}_t$ is the price level in deviation form steady state. Observe that: $\pi_t = \hat{P}_t - \hat{P}_{t-1}$. Is there a value for ϕ_P that allows the central bank to eliminate arbitrary variations in inflation due to sunspots? This leads to the next proposition.

Proposition 5 (Wicksellian Rule)**.** *Suppose monetary policy is set so that* $\hat{\imath}_t = \phi_P \hat{P}_t + \varepsilon_t^i$*. Then, there is a unique bounded rational expectation equilibrium for the price level if* $\phi_p > 0$*.*

To prove this proposition, we start by substituting, as before, the policy rule into the Fisher equation: $\phi_P \hat{P}_t + \varepsilon_t^i = \hat{r}_t + E_t \hat{P}_{t+1} - \hat{P}_t$. Then, solving for the price level and iterating forward gives:

$$\begin{aligned} \hat{P}_t &= \frac{1}{1+\phi_P} E_t \hat{P}_{t+1} + \frac{1}{1+\phi_P} \left(\hat{r}_t - \varepsilon_t^i \right) \\ &= \left(\frac{1}{1+\phi_P} \right)^T E_t \hat{P}_{t+T} + \sum_{j=0}^{T-1} \left(\frac{1}{1+\phi_P} \right)^{j+1} E_t \left(\hat{r}_{t+j} - \varepsilon_t^i \right) \end{aligned}$$

yielding a unique bounded solution:

$$\hat{P}_t = \sum_{j=0}^{\infty} \left(\frac{1}{1+\phi_P} \right)^{j+1} E_t \left(\hat{r}_{t+j} - \varepsilon_t^i \right), \tag{28}$$

if $\left| \frac{1}{1+\phi_P} \right| < 1$, which is satisfied under Wicksell's proposal. Indeed, $\phi_p > 0$ under the interpretation above. Hence, any rule that has the Wicksellian property will lead to determinacy.

3.1.5 Generalizations and Extensions

There are number of generalizations and extensions that can be done and the exact condition for determinacy will differ if one introduces features such as lags in the interest rate rule (so that the interest rate moves more slowly), endogenous output determination, linearization around trend inflation, and so on. The gist of the argument survives, however: the nominal interest rate should increase more that one-to-one, and policy in the 1970s was such that it allowed for self-fulfilling inflation.

An important observation is that the central bank can fully implement price stability if it follows a policy rule of the form

$$\hat{\imath}_t = \hat{r}_t + \phi_\pi \hat{\pi}_t \tag{29}$$

for any $\phi_\pi > 1$. More generally, if the central bank targets the natural rate of interest, which we will define in future chapters, it will implement full price stability.

4 The Optimal Choice of Policy Instrument

We have seen that either a money supply or an interest rate policy rule can bring about a well-defined price level. Is there, then, nothing that can be said about which policy instrument is more desirable? To answer this question, let us suppose that the objective of the government is to minimize the variance of inflation, $Var(\pi_t)$, an objective that will closely resemble social welfare in more complete models.

Here, I want to put some structure on the reason why most central banks use the interest rate rather than the money supply as their operating target. A common argument for monetary targeting is that the money supply is directly controlled by the central bank, and thus arguably easy to monitor. The downside of this approach is that there might be large shocks to money supply. Meanwhile, interest rate targeting relies on a reaction to inflation. The downside of this

approach is that inflation is observed with a noise, and thus that policy rule might be more difficult to implement and monitor. To formalize this let us assume that inflation is only observed with error, so that the policy rule is:

$$\hat{\imath}_t = \hat{r}_t \phi_\pi \pi_t^*,$$

where π_t^* is observed inflation. There is a measurement error in inflation, i.e.:

$$\pi_t = \pi_t^* + \epsilon_t,$$

where, for simplicity, ϵ_t is iid $N(0, \sigma_\epsilon)$ and r_t is iid $N(0, \sigma_r)$. This implies, following the same argument as before, that the unique bounded solution is given by:

$$\pi_t = \epsilon_t$$

Hence, the variance of inflation is given by:

$$Var(\pi_t) = \sigma_\epsilon^2$$

Consider now a policy that targets money supply. One argument for such a rule is that it is easy to monitor and does not need much information, at least if the money supply is interpreted very narrowly. Hence, consider the policy rule:

$$\hat{\mu}_t = 0$$

As we have already seen, if we assume that: $E_t \hat{Y}_{t+j} = E_t \hat{r}_{t+j} = E_t \epsilon_{t+j}^{md} = 0$, then, from (25), we get:

$$\hat{m}_t = \frac{1}{1+\eta_i}(\eta_y \hat{Y}_t - \eta_i \hat{r}_t + \epsilon_t^{md}),$$

From (20):

$$\begin{aligned} \pi_t &= m_{t-1} - m_t \\ &= \frac{1}{1+\eta_i}[\eta_i(\hat{r}_t - r_{t-1}) - \eta_y(Y_t - Y_{t-1}) - (\epsilon_t^{md} - \epsilon_{t-1}^{md})] \end{aligned}$$

so that:

$$Var(\pi_t) = 2\left[\left(\frac{\eta_i}{1+\eta_i}\right)^2 \sigma_r^2 + \left(\frac{\eta_y}{1+\eta_i}\right)^2 \sigma_y^2 + \left(\frac{1}{1+\eta_i}\right)^2 \sigma_{md}^2\right]$$

Which policy arrangement is more desirable? That depends on the implied relative variance of inflation according to each rule. Inspecting the equations above reveals that interest rate is a better policy instrument if:

$$\sigma_\epsilon^2 < 2\left[\left(\frac{\eta_i}{1+\eta_i}\right)^2 \sigma_r^2 + \left(\frac{\eta_y}{1+\eta_i}\right)^2 \sigma_y^2 + \left(\frac{1}{1+\eta_i}\right)^2 \sigma_{md}^2\right] \qquad (30)$$

The key point is that the extent to which monetary targeting works better than interest rate targeting depends crucially on the stability of the money demand function. If the money demand function is very unstable, then a money target will result in volatile inflation. The instability of inflation due to money demand fluctuations is given by the right hand side of (30), while the instability of the measurement error of inflation is given by the left. It was the instability of money demand in the 1980s, i.e. a constant supply of money led to large fluctuation in interest rate, that is commonly given as the reason why the Fed abandoned that strategy.

5 Global Determinacy

So far, we have only considered the question of local determinacy of equilibria. Thus, we only ask if there is determinacy in a small neighborhood of the equilibrium. Here, we consider whether there are other global solutions, assuming that fiscal policy is sustainable. We find that there are, but I will argue that these are fragile and not interesting to study.

Consider the fully non-linear model. For simplicity, I focus on a perfect foresight equilibrium.

We assume, for simplicity, that utility of consumption is given by $\log C_t$ and that the utility of holding real money balances is given by:

$$\chi\left(\frac{M_t}{P_t}\right) = \frac{M_t}{P_t} - \frac{\chi_0}{1+\chi}\left(\frac{M_t}{P_t}\right)^{1+\chi} \quad \text{for } i > 0.$$

This specification for real money balances implies a satiation point at which the marginal utility is 0, i.e. $\chi_m = 0$ when $\frac{M}{P} = \left(\frac{1}{\chi_0}\right)^{\frac{1}{\chi}}$, so that:

$$\chi\left(\frac{M_t}{P_t}\right) = \begin{cases} \left(\frac{1}{\chi_0}\right)^{\frac{1}{\chi}} & \text{if} \quad \frac{M_t}{P_t} \geq \left(\frac{1}{\chi_0}\right)^{\frac{1}{\chi}} \\ \frac{M_t}{P_t} - \frac{\chi_0}{1+\chi}\left(\frac{M_t}{P_t}\right)^{1+\chi} & \text{if} \quad \frac{M_t}{P_t} < \left(\frac{1}{\chi_0}\right)^{\frac{1}{\chi}} \end{cases}$$

We assume that $Y_t = \bar{Y}$ at all times, so the model can be summarized by the following equations:

$$\beta^{-1}\Pi_{t+1} = 1 + i_t \tag{31}$$

$$\chi_0\left(\frac{M_t}{P_t}\right)^{\chi} = \frac{1}{1+i_t} \text{ if } i > 0 \tag{32}$$

$$\frac{M_t}{P_t} \geq \left(\frac{1}{\chi_0}\right)^{\frac{1}{\chi}} \text{ if } i = 0 \tag{33}$$

$$m_t = m_{t-1}\mu_t\Pi_t^{-1} \tag{34}$$

where (31) is the Euler Equation, (32) is the money demand at positive interest rates, (33) the money demand at the zero interest rate and (34) is an identity.

We now specify the non-linear Taylor rule:

$$1 + i_t = \max[1, (1 + \bar{\imath})\Pi_t^{\phi_\pi}]$$

Then, the equilibrium determination can be summarized by a single equation:

$$\Pi_{t+1} = \max\left[\beta, \Pi_t^{\phi_\pi}\right] \tag{35}$$

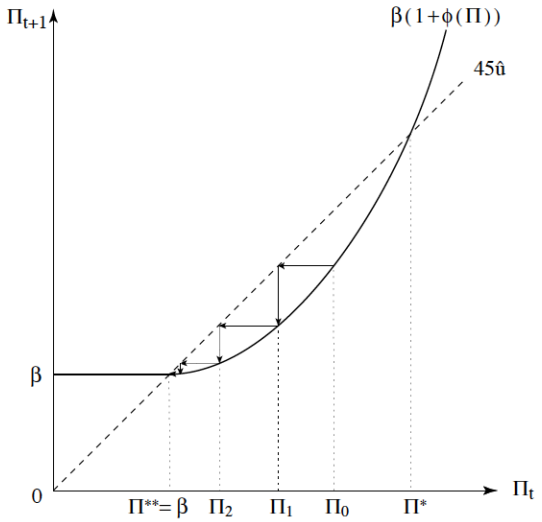


Figure 2.4: A self-fulfilling deflation under a Taylor rule.

Figure 4: Self-fulfilling Deflation

Any path for inflation that satisfies this equation represents an equilibrium. The equilibrium determination is plotted in Figure 4. Starting from any Π_0, we can construct a sequence $\{\Pi_t\}$ that satisfies equation (35) at all dates. There are two possibilities: one is an explosive path for inflation, represented by the right-hand side of the point Π^* in the figure; the other is a deflationary path until the zero lower bound is binding, so that $\Pi_{t+1} = \beta$.

Consider first the explosive path. Suppose that the Taylor coefficient is 1.5, a common value in the literature. Using this value, we can compute the implied path for inflation, starting off, for example, at $\Pi_0 = 1.1$, then $\Pi_1 = (1.1)^{1.5} = 1.15$ and so on. Hence, there is an inflationary spiral consistent with any initial condition $\Pi_0 > 1$. Moving forward in time in such an inflationary spiral (starting

close to $\Pi_0 = 1$), in two and a half years prices double every day, in eight years prices double every second.

It is important to understand how this construction is derived. While I constructed the equilibrium starting from a specific initial value for Π_0 and then deriving Π_1 and Π_2 and so on, this is not the way in which such an equilibrium occurs. Instead, it is the *rational expectation* of Π_1, Π_2, ... constructed in the spiral that triggers Π_0 to be different from Π^* and thus gets the inflation process started. Note that, in order for the inflationary spiral to start, one does not only need to expect Π_1 to be 1.15, but also $\Pi_2 = ((1.1)^{1.5})^{1.5}$ and so on into the infinite. If, at some point, the agents do not expect ever-increasing inflation, the spiral breaks down and Π_0 can only take the value $\Pi_0 = 1$.

Does this set of expectations correspond to a "reasonable equilibrium" and is it plausible that private sector expectations will coordinate on them? One way to exclude the possibility of such an equilibrium is to assume that the expected rate of change of prices can never be above some arbitrarily large number, say Π^{high}, which represents an upper bound of how high inflation can be (for example, because there is a limit on how fast money can be printed due to the finite number of trees!). If such an upper limit exists, this construction unravels and the inflationary path can be excluded. There are a variety of ways in which people have formalized this type of idea, i.e. have given reasons for why there might be an upper bound on how high inflation can go. For example, Obstfeld and Rogoff (1986) *assume* that at high enough inflation the government switches to a commodity standard where the currency has some (perhaps arbitrarily small) ϵ value that is finite. This provides an upper bound on inflation and thus the spiral unravels.

What about the deflationary path? The most natural way to exclude it is by relaxing the assumption of sustainable fiscal policy in ways that look reasonable. They key thing to notice is that in the deflationary path the price level is continuously declining. In defining the equilibrium, we assumed that fiscal policy was sustainable, i.e that the transversality condition, which we can write as $\lim_{T\to\infty} \beta^T \frac{V_T}{P_T} = 0$, was satisfied in every period. For fiscal policy to be sustainable in the way we defined it, the condition means that the government needs to reduce the nominal value of V_t over time. Consider the possibility that the interest rate is zero. Suppose furthermore that the government will never issue debt in negative supply (i.e. $B_t \geq 0$). Now let us suppose that the government will never issue money of a smaller quantity than, say, 1 dollar, i.e. $M_t \geq \epsilon$. Then, one can show that, under this fiscal policy specification, the transversality condition of the representative household is violated in the deflationary equilibrium and, hence, this is not a valid equilibrium. Therefore, if we think that fiscal policy is sustainable except when $i_t = 0$ and that $B_t \geq 0$ and

$M_t \geq \epsilon$ are reasonable restrictions, we can exclude the deflationary equilibrium.

6 References

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7 Notes on Approximation

7.1 Money Demand

We start by approximating equation (16), assuming the following functional forms: $u(C) = \frac{C^{1-\sigma^{-1}}}{1-\sigma^{-1}}$ and $\chi(\frac{M}{P}) = \frac{(\frac{M}{P})^{1-\nu}}{1-\nu}$:

$$\frac{\left(\frac{M_t}{P_t}\right)^{-\nu}}{Y_t^{-\sigma^{-1}}} = \frac{i_t}{(1+i_t)}$$

Defining $\frac{M_t}{P_t} = m_t$, in steady state:

$$\frac{\bar{m}^{-\nu}}{\bar{Y}^{-\sigma^{-1}}} = \frac{\bar{i}}{(1+\bar{i})}$$

Let's take a first-order Taylor approximation around the steady state:

$$\begin{aligned}
-\nu\bar{m}^{-\nu-1}\bar{Y}^{\sigma^{-1}}dm_t + \sigma^{-1}\bar{m}^{-\nu}\bar{Y}^{\sigma^{-1}-1}dY_t &= \frac{1}{(1+\bar{i})}di_t - \frac{\bar{i}}{(1+\bar{i})^2}di_t \\
-\nu\bar{m}^{-\nu}\bar{Y}^{\sigma^{-1}}\frac{dm_t}{\bar{m}} + \sigma^{-1}\bar{m}^{-\nu}\bar{Y}^{\sigma^{-1}}\frac{dY_t}{\bar{Y}} &= \left(1 - \frac{\bar{i}}{(1+\bar{i})}\right)\frac{1}{(1+\bar{i})}di_t \\
-\nu\bar{m}^{-\nu}\bar{Y}^{\sigma^{-1}}\hat{m}_t + \sigma^{-1}\bar{m}^{-\nu}\bar{Y}^{\sigma^{-1}}\hat{Y}_t &= \frac{1}{(1+\bar{i})}\hat{i}_t
\end{aligned}$$

Using the fact that in steady state we have: $\bar{m}^{-\nu}\bar{Y}^{\sigma^{-1}} = \frac{\bar{i}}{(1+\bar{i})}$, we can rearrange and cancel out the terms as follows:

$$\begin{aligned} -\nu\bar{m}^{-\nu}\bar{Y}^{\sigma^{-1}}\hat{m}_t + \sigma^{-1}\bar{m}^{-\nu}\bar{Y}^{\sigma^{-1}}\hat{Y}_t &= \frac{\bar{i}}{(1+\bar{i})}\frac{1}{\bar{i}}\hat{i}_t \\ -\nu\hat{m}_t + \sigma^{-1}\hat{Y}_t &= \frac{1}{\bar{i}}\hat{i}_t \end{aligned}$$

Using again the assumptions above and the fact that $\bar{\Pi} = 1$: $1 + \bar{i} = \beta^{-1} \Longrightarrow \bar{i} = \frac{1-\beta}{\beta}$:

$$-\nu\hat{m}_t + \sigma^{-1}\hat{Y}_t = \frac{\beta}{1-\beta}\hat{i}_t$$

Finally, we get:

$$\hat{m}_t = \nu^{-1}\sigma^{-1}\hat{Y}_t - \nu^{-1}\frac{\beta}{1-\beta}\hat{i}_t$$

7.2 Consumption Euler Equation

Let's continue with equation (15):

$$\frac{1}{(1+i_t)} = \beta E_t \frac{Y_{t+1}^{-\sigma^{-1}}}{Y_t^{-\sigma^{-1}}}\frac{\xi_{t+1}}{\xi_t}\underbrace{\frac{P_t}{P_{t+1}}}_{\Pi_{t+1}^{-1}}$$

Take a first-order Taylor approximation around the steady state:

$$\begin{aligned} -\frac{1}{(1+\bar{i})^2}di_t = -\beta\sigma^{-1}\frac{\bar{Y}^{-\sigma^{-1}-1}}{\bar{Y}^{-\sigma^{-1}}}E_t dY_{t+1} + \beta\sigma^{-1}\frac{\bar{Y}^{-\sigma^{-1}}}{\bar{Y}^{-\sigma^{-1}+1}}dY_t + \beta\frac{1}{\bar{\xi}}E_t d\xi_{t+1} \\ - \beta\frac{\bar{\xi}}{\bar{\xi}^2}d\xi_t - \beta\frac{1}{\bar{\Pi}^{-2}}E_t d\Pi_{t+1} \end{aligned}$$

Evaluating the expansion at the steady state, which takes the form $\frac{1}{1+\bar{i}} = \beta$:

$$\begin{aligned} -\beta\hat{i} &= \beta\left[-\sigma^{-1}E_t\hat{Y}_{t+1} + \sigma^{-1}\hat{Y}_t + E_t\hat{\xi}_{t+1} - \hat{\xi}_t - E_t\pi_{t+1}\right] \\ -\hat{i} &= -\sigma^{-1}E_t\hat{Y}_{t+1} + \sigma^{-1}\hat{Y}_t + E_t\hat{\xi}_{t+1} - \hat{\xi}_t - E_t\pi_{t+1} \\ \sigma^{-1}\hat{Y}_t &= \sigma^{-1}E_t\hat{Y}_{t+1} - \hat{\imath}_t + E_t\pi_{t+1} + \hat{\xi}_t - E_t\hat{\xi}_{t+1} \end{aligned}$$

Finally, we get:

$$\hat{Y}_t = E_t\hat{Y}_{t+1} - \sigma\left(\hat{\imath}_t - E_t\pi_{t+1} - \hat{\xi}_t + E_t\hat{\xi}_{t+1}\right)$$

7.3 Identity

Finally, the last equation (17):

$$m_t = m_{t-1}\mu_t\Pi_t^{-1}$$

Let's take a first-order Taylor approximation around the steady state, which is $\bar{\mu} = \bar{\Pi} \Longrightarrow \bar{\mu}\bar{\Pi}^{-1} = 1$:

$$\begin{aligned} dm_t &= \bar{\mu}\bar{\Pi}^{-1}dm_{t-1} + \bar{m}\bar{\Pi}^{-1}d\mu_t - \bar{m}\bar{\mu}\bar{\Pi}^{-2}d\Pi_t \\ \frac{\bar{m}}{\bar{m}}dm_t &= \frac{\bar{m}}{\bar{m}}dm_{t-1} + \bar{m}\bar{\Pi}^{-1}\frac{\bar{\mu}}{\bar{\mu}}d\mu_t - \bar{m}\bar{\Pi}^{-1}d\Pi_t \\ \bar{m}\hat{m}_t &= \bar{m}\hat{m}_{t-1} + \bar{m}\hat{\mu}_t - \bar{m}\pi_t \end{aligned}$$

Finally, we get:

$$\hat{m}_t = \hat{m}_{t-1} + \hat{\mu}_t - \pi_t$$

8 Problems

Problem 1
Consider this model:

$$\begin{aligned} \hat{\imath}_t &= \hat{r}_t + E_t\pi_{t+1} \\ \hat{m}_t &= \eta_y\hat{Y}_t - \eta_i\hat{\imath}_t \\ \hat{m}_t &= \hat{m}_{t-1} + \hat{\mu}_t - \pi_t \\ \hat{\mu}_t &= \gamma\hat{\mu}_{t-1} + \epsilon_t \\ \hat{r}_t &= \rho\hat{r}_{t-1} + \upsilon_t \end{aligned}$$

where ϵ_t and υ_t are iid with zero mean, $0 < \rho$ and $\gamma < 1$. Here, the notation is the same as in class: i is interest rate, m is real money balance, π is inflation, Y is output, r is an exogenous process for the real interest rate and μ is the nominal growth rate of money. The variable $\hat{Y}_t$ is an exogenous endowment, which is iid with zero mean.

i. Solve for inflation as a function of current and expected future values of $\hat{\mu}_t$, $\hat{r}_t$ and $\hat{Y}_t$. Show that there is a unique bounded solution. ***Hint:*** it is useful here to write the system in terms of a single forward-looking equation of $\hat{m}_t$ first and then use the third equation to solve for inflation, given the evolution already derived for $\hat{m}_t$.

ii. Solve for inflation and the nominal interest rate as functions of only current and/or past values of $\hat{\mu}_t$, $\hat{r}_t, \hat{Y}_t$. ***Hint:*** use the last two equations and the fact that $E_t\hat{Y}_{t+j} = 0$ for $j > 0$ to solve for the expectation of $\hat{\mu}_t$ and $\hat{r}_t$ from your last answer.

iii. Show an impulse response function for π_t and $\hat{\imath}_t$ to ϵ_t, i.e. to an increase in money growth.

iv. How well do you think your answer to (iii) fits with empirical evidence? Does this model imply any effect on output? Does the impact on inflation look reasonable?